

SAMEER MAYURESH PATWARDHAN

B.Tech Automation & Robotics

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- Portfolio: <https://sam-airobotics.github.io/portfolio/>

SUMMARY

Automation & Robotics student with hands-on experience in ROS 2, robot simulation, and robotics software. Interested in building reliable autonomous and humanoid robotic systems.

EDUCATION

B.Tech — Automation & Robotics

JSPM's Rajarshi Shahu College of Engineering, Pune, Maharashtra, India

EXPERIENCE

Team CiPHER | Robotics Team

Mechanical Team Member — Jan 2025 – Jul 2026

- Worked on mechanical design and robot-system development as part of a multidisciplinary robotics team.
- Supported mechanical integration and collaborated with teammates to turn robot concepts into practical assemblies.

Programming Team Member — Jan 2026 – Present

- Develop and test robotics software, simulation workflows, and system integrations.
- Use Linux, Git, GitHub, ROS 2, and simulation tools to build, debug, and iterate on robotics projects.

PROJECTS

FusionToDescription — Built a Fusion/CAD-to-robot-description workflow using URDF/Xacro, with simulation-oriented output.

Cortex - Hospital Assistance Robot — Worked on a mobile/half-humanoid robot concept for hospital medicine transport and verification, with focus on autonomous robotics, sensing, and human-robot interaction.

Robotic Arm Simulation — Developed a simulated robotic arm setup covering robot description, joint configuration, RViz visualization, and Gazebo simulation.

AMR Simulation — Built an Autonomous Mobile Robot simulation workflow covering robot motion, sensor-based perception, visualization, and navigation-oriented behavior.

TECHNICAL SKILLS

ROS & Simulation	ROS 2 Jazzy, Gazebo Harmonic, RViz, Xacro, URDF, robot simulation
Programming	C++, Python
Robotics	Sensor Fusion, robot modeling, kinematics, autonomous robotics, mobile robots
Development Tools	Linux, Git, GitHub, debugging, testing, version control

RELEVANT AREAS

Robot Control • Kinematics • State Estimation • Sensor Integration • LiDAR • Cameras • IMUs • Depth Sensors • Encoders • Autonomous Mobile Robots • Humanoid Robotics • Sim-to-Real

RESEARCH FOCUS

Bionics • Humanoid Robotics • Autonomous Systems • Robot Perception • Sensor Fusion